WORKED SOLUTION

STEP II 2002, Question 10

Mechanics • Kinematics and Optimisation

STEP II 2002, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s2-q10

The key idea

The total distance is fixed, so the race time T and the switching time t are tied together by a single equation. Clearing the fraction turns it into a quadratic in t, and a time T is achievable only if that quadratic has a real root. The discriminant condition gives the minimum T directly, with no calculus. For the second part, once we know both runners' positions as functions of time, the distance between them is a piecewise quadratic, and we maximise it in the usual way.

Solution

Throughout, distances are in km and times in hours, and $42\frac{3}{8} = \frac{339}{8}$.

The minimum race time

The first competitor covers $13t$ in the first t hours and then $(14 + \frac{2t}{T})(T - t)$ in the remaining $T - t$ hours. So

$$13t + \left(14 + \frac{2t}{T}\right)(T - t) = \frac{339}{8}.$$

Expanding, $13t + 14T - 14t + 2t - \dfrac{2t^2}{T} = \dfrac{339}{8}$, that is

$$t + 14T - \frac{2t^2}{T} = \frac{339}{8}.$$

Multiplying by $T > 0$ and collecting powers of t:

$$2t^2 - Tt + \left(\frac{339}{8}T - 14T^2\right) = 0. \qquad (*)$$

Ask yourself We want the smallest T, but T is not given as a function of t. How can we tell which values of T are possible at all?

A value of T can occur only if $(*)$ has a real solution t, so its discriminant must be non-negative:

$$T^2 - 8\Big(\frac{339}{8}T - 14T^2\Big) \geq 0 \iff 113T^2 - 339T \geq 0 \iff 113T(T-3) \geq 0.$$

Since $T > 0$, this forces $T \geq 3$.

The bound is attained. When $T = 3$ the discriminant is zero and $(*)$ becomes

$$2t^2 - 3t + \Big(\frac{1017}{8} - 126\Big) = 2t^2 - 3t + \frac{9}{8} = 2\Big(t - \frac{3}{4}\Big)^2 = 0,$$

so $t = \frac{3}{4}$. This lies between 0 and T, so it is a genuine race plan: 45 minutes at 13, then the remaining $2\frac{1}{4}$ hours at $14 + \frac{2(3/4)}{3} = 14.5$. Indeed $13 \cdot \frac{3}{4} + 14.5 \cdot \frac{9}{4} = 9.75 + 32.625 = 42.375$. Hence the minimum possible value of T is 3, as required.

(Equivalently, differentiating $(*)$ implicitly and setting $\frac{\mathrm{d}T}{\mathrm{d}t} = 0$ gives $T = 4t$, and substituting back gives $T = 3$. The discriminant argument has the advantage of proving that 3 is a global minimum, not just a stationary value.)

The maximum separation

Let τ be the time since the start. The second competitor's speed is $16 - k\tau$ for some constant k, so their distance run is

$$s_2(\tau) = 16\tau - \tfrac{1}{2}k\tau^2.$$

Finishing in 3 hours gives $48 - \frac{9}{2}k = \frac{339}{8}$, so $\frac{9}{2}k = \frac{45}{8}$ and $k = \frac{5}{4}$. Thus $s_2 = 16\tau - \frac{5}{8}\tau^2$, and the speed falls from 16 to 12.25, staying positive.

The first competitor also takes 3 hours, which by the first part means $t = \frac{3}{4}$. Their distance run is

$$s_1(\tau) = \begin{cases} 13\tau, & 0 \leq \tau \leq \frac{3}{4}, \\ \frac{39}{4} + \frac{29}{2}(\tau - \frac{3}{4}), & \frac{3}{4} \leq \tau \leq 3. \end{cases}$$

Let $d = s_2 - s_1$. The second runner starts faster, so we expect $d \geq 0$.

For $0 \leq \tau \leq \frac{3}{4}$: $d = 3\tau - \frac{5}{8}\tau^2$, with $\frac{\mathrm{d}d}{\mathrm{d}\tau} = 3 - \frac{5}{4}\tau > 0$. So d increases on this interval, reaching $d(\frac{3}{4}) = \frac{9}{4} - \frac{45}{128} = \frac{243}{128}$.

For $\frac{3}{4} \leq \tau \leq 3$:

$$d = 16\tau - \tfrac{5}{8}\tau^2 - \tfrac{39}{4} - \tfrac{29}{2}\tau + \tfrac{87}{8} = \tfrac{1}{8}(9 + 12\tau - 5\tau^2) = \tfrac{1}{8}(3-\tau)(3+5\tau).$$

This is a downward parabola with vertex at $\tau = \frac{12}{10} = \frac{6}{5}$, which lies inside $[\frac{3}{4}, 3]$. (Equivalently, the gap stops growing when the speeds are equal: $16 - \frac{5}{4}\tau = 14.5$.) There

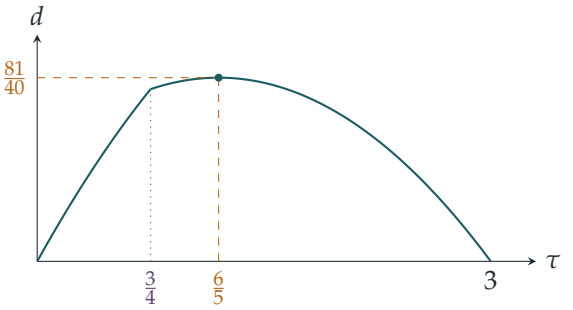
$$d(\tfrac{6}{5}) = \tfrac{1}{8} \cdot \tfrac{9}{5} \cdot 9 = \tfrac{81}{40}.$$

The factorised form shows $d \geq 0$ on the whole interval, falling to 0 at the finish, as it must.

Comparing the two pieces, $\frac{81}{40} = 2.025 > \frac{243}{128} \approx 1.898$. So the maximum distance between the runners is

$$\boxed{\tfrac{81}{40}\text{ km} = 2.025\text{ km}},$$

reached 1.2 hours (72 minutes) into the race, with the second competitor ahead.



Checks

- Both runners finish at $\tau = 3$, and the formula gives $d(3) = 0$, as it must.
- The maximum of the second piece occurs exactly when the speeds are equal (14.5 each), which is the natural condition for a gap to stop growing.
- Numerically: solving the quadratic for T over a fine grid of t gives the smallest T as 3.000 at $t = 0.75$, and evaluating d at 300 000 times in $[0, 3]$ gives a maximum of 2.025 at $\tau = 1.2$, with $d \geq 0$ throughout.

Common slips

- Using the first competitor's minimum-time plan without noticing that "run time of 3 hours" is precisely what fixes $t = \frac{3}{4}$.
- Maximising only at the kink $\tau = \frac{3}{4}$: the gap keeps growing after the switch, because the second runner is still faster than 14.5 until $\tau = \frac{6}{5}$.
- Writing the second runner's distance as $16\tau - k\tau^2$, forgetting the $\frac{1}{2}$ from integrating a linearly decreasing speed.

Beyond STEP

The first part is an optimisation over a constraint given implicitly by $F(t, T) = 0$. Asking "for which T does a solution t exist?" is the idea behind the implicit function theorem and, more generally, Lagrange multipliers: at the optimum the level curve of the

constraint has $\partial F/\partial t = 0$, which is exactly the double-root condition here. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats kinematics with calculus throughout, and Siklos, *Advanced Problems in Mathematics*, has more STEP-style optimisation.

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